Relational Gains, Privacy Strains: Exploring Users' Perceptions and Experiences with ChatGPT's Memory Feature

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Abstract

ChatGPT's memory feature is designed to provide users with greater control and more helpful responses. Yet, it remains unclear how users perceive this feature in relation to privacy. To address this gap, we conducted interviews with 20 ChatGPT users from diverse backgrounds. Our findings revealed four major characteristics that distinguish ChatGPT's memory from human memory: perceived unforgetfulness, detailedness, accuracy, and lack of emotions, highlighting the machine-like nature of AI memory. Moreover, both ChatGPT's memory and human memory were perceived as beneficial for relationship building. Notably, most participants experienced negative expectancy violations after learning what ChatGPT remembered about them. They expressed a strong need for greater visibility, accessibility, transparency, and user control in the design of future memory features. Drawing on users' suggestions and theoretical frameworks on privacy management, we provide design implications for developing a more transparent, responsible, and user-aligned memory experience that helps them navigate privacy-personalization trade-offs when interacting with LLM-based memories.

CCS Concepts

• **Human-centered computing → Empirical studies in HCI**.

Keywords

ChatGPT, Memory Feature, User Perception, AI Memory, Unforgetfulness, User Control, Transparency

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1 Introduction

Memories are essential to human experience, shaping how we learn, adapt, and understand the world. Generative AI systems, such as ChatGPT [49] and Claude [4], have introduced memory features that enable them to retain and accumulate information across conversations to provide more personalized interactions to their users. Since ChatGPT is one of the most widely used generative AI tools in the U.S. [1] and was among the first to release its user-facing memory feature [49], this study focuses primarily on users' perceptions and experiences with ChatGPT's memory feature.

According to OpenAI, ChatGPT's memory feature was designed to enhance user control and provide more personalized chats [49]. Users can explicitly ask ChatGPT to remember details, such as their names, favorite color, or dietary preferences, while the system also automatically retains and applies information from previous chats to improve future interactions [51]. Although memory functions as an internal mechanism within large language models (LLMs), ChatGPT may display visual cues, such as "Memory Updated" or "Updated Saved Memory" (see Figure 1), to signal when new information has been stored. Users can also access a list of memories through the system settings and choose to delete or remove them.

Notably, ChatGPT's memory feature relies on continuously monitoring users' chat histories to extract information that might be relevant for future conversations. Since the operation of the memory feature is often opaque, users may develop their own mental models or folk theories to understand how the feature works. These understandings evolve over time, particularly when expectations are violated in practice [23]. We seek to document these moments where users experience shifts in their mental models, particularly after discovering what ChatGPT remembers about them.

While ChatGPT's memory feature is designed with good intentions, prior research shows that the current design of LLMs can limit users' awareness and comprehension of privacy risks, sometimes even encouraging sensitive information disclosures [77]. Furthermore, researchers have also identified risks associated with AI memorization [33, 57]. For example, LLMs may store users' data

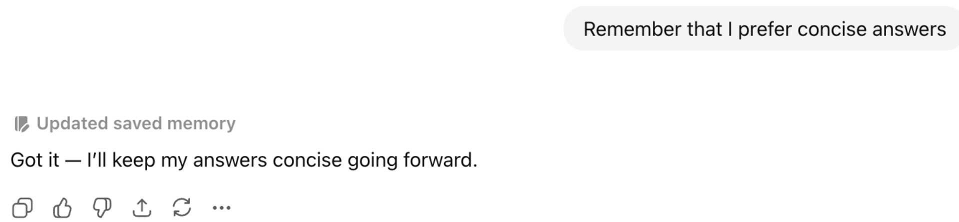

Figure 1: ChatGPT Responded with "Updated saved memory"

for future model training without explicit consent [57], infer sensitive information about users from seemingly innocent inputs [57], or reproduce portions of users' data when prompted by adversaries [33, 77]. When LLMs are integrated into external agents or third-party modules, privacy concerns become even more complex, as these systems may generate, access, and disseminate sensitive user data across interconnected components with varying levels of security and commitment to privacy protections [57].

As generative AI systems begin to retain information across prolonged sessions and various contexts, persistent AI memory also introduces new privacy concerns related to the construction of user profiles [66]. Although theoretical frameworks such as privacy calculus [37] and calls for trustworthy and responsible AI [41] offer guidance for improving the transparency and responsibility of LLM-based generative AI systems, it remains unclear which privacy-related needs should be prioritized when designing ChatGPT's memory feature. Drawing on a user-centered approach, we examine the privacy concerns that arise in interactions with ChatGPT's memory feature. By translating these concerns into actionable design suggestions, this study aims to improve the current design of ChatGPT's memory feature to better support privacy protection and digital well-being.

Together, we are guided by the following three research questions:

- RQ1: How do users perceive ChatGPT's memory in comparison to human memory?
- RQ2: How do users' mental models change after learning what ChatGPT remembers about them?
- RQ3: What suggestions do users offer to enhance ChatGPT's memory feature so that it better safeguards their privacy?

To answer these research questions, we conducted semi-structured interviews with 20 users to explore their perceptions and experience with ChatGPT's memory feature. We designed an activity that prompted participants to examine what ChatGPT remembered about them. This approach allowed us to observe how their understanding evolved as they learn more about the feature. Additionally, we documented instances where ChatGPT's memory feature aligned with or fell short of their expectations. These cases help uncover the sources of user satisfaction or frustration and shed light on users' mental models of the feature.

By examining users' perceptions and lived experiences with ChatGPT's memory feature, particularly in relation to privacy, this study makes three key contributions. First, it extends emerging work on computational memory mechanisms for LLM agents [30, 39, 42, 43, 64, 72, 78] by foregrounding users' real-world interactions with ChatGPT's memory feature. Understanding users' lived experiences with the feature is crucial for revealing both the perceived relational benefits and the privacy risks that accompany persistent AI memory. Second, it empirically documents moments where ChatGPT's memory feature was experienced as satisfying, surprising, or frustrating. These accounts offer concrete opportunities for designing ChatGPT's memory feature that better aligns with users' needs, expectations, and privacy boundaries. Third, by drawing on and extending insights from these user experiences, this study advances theoretical understandings of AI memory as an affordance [21], i.e., it situates at the intersection of an objective interface capability and users' subjective perceptions of it. The ability of LLM-based memories to retain personal information with high accuracy and detailedness, and to use this information to support relationship-building has profound implications for human cognition, affect, and behavior, especially as users engage with ChatGPT and similar LLMs over extended periods.

2 Related Work

In this section, we first examine the similarities and differences between human memory and AI memory. We then review existing research on users' mental models, focusing on how people understand the memory feature and how expectancy violations can prompt updates to these models. Finally, we address privacy as a key concern related to AI memorization, drawing on concepts from privacy calculus and privacy personalization paradox to guide strategies for managing users' privacy when interacting with ChatGPT's memory feature.

2.1 Human Memory and AI Memory

Memory is fundamental to human experience, which refers to the brain's ability to store and retrieve information. Budson and Kensinger [9] described memory as a set of interconnected abilities in that multiple memory systems work together, encoding information across different regions of the brain. Similar to human memories, researchers have pointed out three types of AI memories that collectively operate within ChatGPT [76]. Contextual memories keep track of conversation history only within a single session, making it relatively short-lived. Retrieval-augmented generation

(RAG)-based memories, on the other hand, extract and store user information across sessions, building longer-term user profiles that the system can draw upon later. Meanwhile, epistemological memories consist of stored facts and learned knowledge, blurring the boundary between personal data and general knowledge. Informed by the categories of human memory, including short-term (sensory and working memory) and long-term (explicit and implicit memory), Wu et al. [69] introduced a novel classification method for AI memory based on three dimensions: object (personal vs. system memory), form (parametric vs. non-parametric memory), and time (short-term vs. long-term memory).

When comparing the performance of human memory and AI memory, previous research has identified both similarities and differences. For example, Cao et al. [11] found that both AI memory and human memory store less when overloaded with information and store more with repeated exposure. They also face comparable challenges when retrieving overlapping content, where too many similar facts create confusion. Like human memory, AI memory is vulnerable to false memories, i.e., recalling related but unseen items, and can extend prior learning to new, related contexts through cross-domain generalization. Likewise, Janik [31] identified surprising similarities between AI memory and key characteristics of human memory. He argued that these similarities likely stem from LLMs learning from human-generated text, since "the biological features of human memory leave an imprint on the way that we structure our textual narratives" [31, p. 1].

Despite the similarities, Cao et al. [11] pointed out that AI memory differs from human memory in two major ways: they are less influenced by the positional bias, i.e., the order in which information is presented, and more robust when processing random or meaningless material. Rolls [53] also highlighted the functional difference between AI memory and human memory, in that humans store and recall memories using local associations in the brain, allowing them to retrieve whole, context-rich experiences from partial cues. These memories include details about what is seen, where objects and people are, and the surrounding context, and they support planning and imagining future actions. In contrast, generative AI models like ChatGPT generate text one piece at a time using global error-driven learning; they do not store or recall full experiences and lack the contextual and flexible memory capabilities of the human brain.

Building on this distinction, Hoskins [29, p. 17] described human memories as "an active, willed, functional, deliberating memory, seen as cognitive and as fundamentally part of human identity, that evolves with time and context in and of the present." This definition emphasizes human agency in deciding which information to retain, in making sense of stored information, and in further shaping identities. However, if people do not attribute mental states to AI, such as goals, intentions, and beliefs [16], then AI memory functions merely as a passive system that codes and retrieves information without agency. Hoskins [29] further explained, humans can re-imagine, remark, and revise past events through the recall process. In his words, memory is "not as a fixed or static entity, but rather as an active process, whereby the past is reconstructed in the present" [29, p. 2]. Similarly, AI can shape new realities by recall [29], which may not necessarily align with the realities experienced by humans. For example, Manovich [44] found that when using AI to re-imagine the urban environment from his childhood, the AI-generated image created representations that were both familiar and unreliable.

On the flip side of remembering, forgetting is defined as "increased episodes of prospective memory loss with delayed processing and response time in aging adults, offset by focus and limiting distraction" [6, p. 1411]. While forgetting is often associated with cognitive challenges and impairment, research in cognitive psychology shows that "forgetting helps people to be happy, well-structured, and context sensitive" [48, p. 551]. However, it is unknown whether AI can truly forget, even though the memory feature allows users to delete or clear ChatGPT's memory.

2.2 Users' Responses to Memory Features in LLM Agents

Extending theoretical discussions on the similarities and differences between AI memory and human memory, empirical research has examined users' awareness and perception of memory features in LLM agents, the types of information these systems retain, and the risks and concerns related to AI memorization. For example, in a survey-based study, Sadat et al. [54] found that about half of their respondents were either unaware of or unsure about the existence of ChatGPT's memory feature. Among those lacking prior awareness, many reported higher discomfort with ChatGPT's automatic activation of the memory feature without prior notification. Only a small proportion of the participants knew how to check the data stored in their account and what ChatGPT had memorized about them.

Sadat et al. [54] also analyzed user-provided AI memories from their ChatGPT accounts, finding that the system primarily stored information related to users' interests, education, employment, and preferences. However, it also retained a small proportion of moderate- or high-risk information, including personal and social details, health, financial, and contact information. Their analysis also revealed vulnerabilities in the system's ability to protect sensitive data as well as inaccuracies in how users were represented in ChatGPT's memory.

2.3 Mental Models of LLM-based Memory and Expectancy Violations

Despite a low level of awareness and knowledge about ChatGPT's memory feature, users can gradually make sense of the feature through interactions and form mental models that guide their understanding and future use of the feature. Conceptually, mental models are psychological representations of real, hypothetical, or imaginary situations. They are "smaller-scale models" of reality that are used to anticipate events, reason, and formulate explanations [14]. This mental model construction process is also commonly referred to as an "imperfect representation", which could contain human errors [46]. Examining users' mental models is especially valuable for understanding how they perceive memory features in LLM agents, whose operations are inherently opaque and difficult to explain [68]. When AI systems provide limited or no explanations, users tend to construct their own rich mental narratives or "algorithmic imaginaries" [8] based on subjective assessments and interpretations. However, these imaginations or psychological representations can be inaccurate or oversimplified compared to the actual functioning of LLMs [34]. For example, Zhang et al. [76] identified three mental models that users form about AI memory across multiple LLM agents: (1) a transient, dialogue-specific buffer, (2) an extension of the model's core training data, and (3) an active information processing mechanism. They also observed an additional mental model reflecting participants' general lack of understanding of how LLM memory functions.

In reality, mental models are not rigid; they are malleable and evolve based on users' expectations and experiences [32]. This adaptability becomes particularly evident when users' expectations are not met, leading them to reassess and reconstruct their understanding of a system [23]. According to the theory of expectancy violations [10], unmet expectations are often evaluated based on their valence, i.e., whether the outcome is perceived as positive or negative relative to the original expectation. Compared to expectancy confirmations, both positive and negative expectation violations have stronger effects on users' evaluations and perceptions. As Burgoon [10, p. 40] explains, "positive violations… are theorized to produce more positive interaction patterns and outcomes than conformity to expectancies; negative violations… are theorized to be detrimental, relative to expectancy confirmation." Prior research further shows that users are more likely to update their mental models when their expectations of AI are violated rather than confirmed [23].

Although users continuously update their mental models of LLM-based memories, there is no guarantee that their updated models are more accurate, as many still lack a full understanding of how LLMs actually work. This is why researchers have argued for the importance of "opening the black box" to help users better calibrate their mental models. Specifically, Liao and Vaughan [41] emphasized the need to increase LLM transparency, enabling stakeholders to better understand the system's capabilities and limitations, the underlying logic of its operations, and how to effectively use or manage its outputs.

2.4 Memorization Risk and Privacy Management

Lack of transparency in LLM-based memory can increase privacy risks, particularly related to what the system memorizes. For example, Zhang et al. [77] documented instances where ChatGPT revealed personal information of a public figure, including name, address, and phone number, to a random user. In addition to concerns about reproducing users' data, previous research shows that these models can also infer sensitive attributes about users from seemingly benign input [57]. Privacy risks become even more complex when LLMs are integrated into third-party tools or modules, as these systems can operate with varying levels of safety, security, and data protection [57]. These AI memorization risks underscore the need for better designs that enhance user control and privacy protection by better data management of the personal information shared, processed, and saved in generative AI systems.

In the privacy literature, privacy calculus has been widely adopted as a theoretical framework to understand user incentives and trade-offs between the expected benefits and risks of disclosing information online. As an earlier example, Culnan and Armstrong [15] applied such human calculus behaviors into online transactions, and found that when companies explicitly inform fair privacy procedures, users are more likely to disclose their personal information based on reduced risks. Later on, Dinev and Hart [17] proposed an extended privacy calculus model in e-commerce and incorporated the elements of trust and interest in the system that can counterbalance perceived privacy risks and promote willingness to disclose personal information. Privacy calculus has been applied to other online venues, including location-based services [71] and mobile disclosures [65]. In general, web-based privacy calculus studies focus on identifying systematic approaches or individual perceptual factors that affect the balance between perceived privacy risks and benefits of sharing personal information.

From the perspective of service providers, the goal becomes to identify a method that results in a positive balance within the benefit-risk equation. One challenge in the process is that greater personalization (as a benefit) tends to paradoxically increase users' sense of vulnerability (as a corresponding cost, simultaneously). Such dilemma is represented by the concept of the personalization privacy paradox, where Awad and Krishnan [5] found that users who rated higher importance of information transparency in web services are less willing to be profiled for personalization. Yet, supporting prevalent e-commerce approaches of making privacy policies and third-party seals more salient, corporate transparency measures are found to enhance service evaluations [19]. Not only the reputation of the service provider but also the completeness/extensiveness of the privacy information can reduce concerns over self-disclosure [3]. Overt (vs. covert) data collection for personalization can increase user engagement with online ads [2], whereas unsolicited push-based (vs. pull-based) location advertising can undermine the effectiveness of corporate seals and policy information in reducing perceived privacy risks [71]. Such positive transparency effects of overt personalization can be explained by enhanced perceived control [13]. Thus, interface features that strengthen transparency and user control can help address the personalization–privacy paradox while maintaining engagement.

More relevant to this study context, Gumusel [25] conducted a literature review of privacy research focused on conversational text-based AI like ChatGPT-3, suggesting that specific AI chatbot features can elicit more data sharing from users. For instance, Kim et al. [35] found that privacy calculus is only activated when an AI chatbot offers personalized (vs. non-personalized) advertising among those with low (vs. high) privacy concerns and future-driven promotion-focused (vs. prevention-focused) orientations. Perceived intellectual competencies of an AI chatbot (i.e., anticipating problems and planning ahead) increase users' willingness to disclose and reduce privacy concerns [22]. Chatbot's ambidexterity (to perform conflicting goals simultaneously) in sales and service offerings could enhance customer experience and patronage [20]. However, it is difficult to identify a specific chatbot approach that has universal effects on disclosure for various interaction contexts.

Alternatively, informing users about data policies can serve as a more straightforward option to reduce privacy concerns and increase trust among AI chatbot users. However, the effectiveness of AI transparency has shown mixed findings. On the one hand, Xu et al. [73] found that transparency explanations of how the chatbot was trained led users to express greater affinity to the chatbot, as well as rate its social intelligence higher. On the other hand, the findings of Sannon et al. [56] suggests that disclosing data sharing practices (with advertisers) increases risk perceptions and lowers the chatbot's trustworthiness, regardless of how socially intelligent the chatbot was perceived. Furthermore, achieving transparency in AI systems can be more challenging due to their foundation on complex black-box models. In response, scholars are striving for explainable AI (XAI) solutions to develop more explainable models, while implementing better literacy and trust measures [18]. General XAI methods can also apply to conversational user interfaces to increase user literacy and engagement, while the conversational nature of LLM interfaces can amplify users' overreliance on the AI system [27]. Thus, identifying effective transparency features or messages within LLMs requires more academic attention. Mindful of the limited literature on LLM memorization in regards to XAI, this study attempts to fill in the research gap by understanding how users assess the benefits and risks of using ChatGPT's memory feature, and obtain preliminary feedback on preferred design interventions to resolve the benefit-risk imbalance or the personalization-convenience paradox.

3 Method

We conducted semi-structured interviews with a total of 20 participants. The sample size meets the recommended saturation benchmark for qualitative studies [24, 28]. The study was approved by the Institutional Review Board (IRB) at Elon university [1] and Michigan State University, and it was preregistered on the Open Science Foundation (OSF)'s website before data collection. Details of the preregistration are available at this link: https://osf.io/5v4y8/overview.

3.1 Participant Screening and Pre-interview Questionnaire

We recruited participants from Prolific, targeting those who were aware of ChatGPT's memory feature. Participants who were interested in the study were directed to a link to Calendly, where they selected their preferred interview date and provided their email address for follow-up communications. The study team contacted them and sent a pre-interview questionnaire to obtain their consent to participate.

To confirm each participant's qualifications, we included screening questions related to prior experience with ChatGPT (i.e., "How often have you used ChatGPT in the past six months?"; 1 = "not at all" to 6 = "used several times each day") and its memory feature (i.e., "To what extent are you aware of ChatGPT's memory feature?"; 1 = "none at all" and 5 = "a great deal"). Those who answered "not at all" or "none at all" to either of the two questions were excluded from the study.

To obtain further ChatGPT usage behaviors, we asked qualified participants to report their frequency of using ChatGPT in both professional and personal use settings across various tasks, including text generation, image generation, music generation, video generation, and others (participants can specify) on a 5-point agreement-based Likert scale. They were also asked to report their prompt literacy with five statements adapted from previous research [38] on a 7-point scale (1 = "strongly disagree" and 7 = "strongly agree"). Sample items included "I can ask appropriate and effective questions to generative AI" and "I can communicate effectively with generative AI." Similarly, we evaluated their perceived AI knowledge by asking their level of agreement with five items adapted from previous research [74], revised to fit this study context. Examples include: "In general, I know a lot about generative AI" and "I understand how generative AI works."

Lastly, all participants provided their demographics, including gender, age, race/ethnicity, educational level, and annual family income. Only participants who completed the consent form, along with the pre-interview questionnaire, were invited to the interview session.

3.2 Participants' Profiles

Demographically, our sample consisted of slightly more males (n = 12) than females (n = 8). Their ages ranged from 19 to 54 years (M = 38, SD = 10.51). They reported using generative AI tools more frequently for personal purposes (M= 3.80, SD = 1.47) than for professional purposes (M = 3.25, SD = 1.71). In addition, their perceived AI knowledge was moderately high (M = 4.99, SD = 1.12), so did the perceived prompting literacy (M = 6.15, SD = .61). We present their profiles in Table 1.

3.3 Interview Protocol

Each interviewer opened with a self-introduction, followed by an explanation of the study's purpose and procedures. Then, permission was asked and obtained to record the Zoom interview session. Participants were allowed to turn off the camera if they felt uncomfortable revealing their physical appearance and surroundings. Following this, participants were asked to briefly introduce themselves, including their preferred name, location, and occupation.

The main interview started with general questions about participants' ChatGPT use, such as their motivations for using the system and common use scenarios. We then asked participants whether they had shared any personal information with ChatGPT, their perceptions of how ChatGPT manages, stores, and uses their personal data, and if they manage their private information on the platform.

Next, we introduced ChatGPT's new feature, Memory, using materials from OpenAI's official website [49]. We mentioned that the memory feature allows ChatGPT to remember user-provided information in order to enhance future interactions. Participants were shown a screenshot illustrating ChatGPT responding with the "Memory Updated" cue after a user shared personal details and were invited to share their likes and dislikes, as well as their expectations about the feature.

The participants were then invited to experiment directly with ChatGPT by entering the prompt "What do you know about me?" They were encouraged to use their own ChatGPT account and share the responses at their comfort level. We asked a few questions to understand their perceptions of the memory feature, such as "Was the result surprising to you?" and "What information would you like to keep stored or delete from ChatGPT?" As we discussed ChatGPT's memory feature, we briefly explained the concept of human memory and invited participants to reflect on the concept of AI memory at a higher-level, with questions such as: "What does it mean to you when AI has memory?" and "How would you compare

[1]The first author's previous university.

Table 1: Participants' Profiles

No.	Gender	Age	Race	Education	Occupation
P1	Male	34	White	Bachelor's degree	Student in CS; self-employed in logistics
P2	Female	37	Black/African American	Bachelor's degree	HR business partner
P3	Female	41	Hispanic	Graduate degree	Salesforce administrator
P4	Female	48	Asian	Graduate degree	Part-time software developer
P5	Male	26	Black or African American	Bechelor's degree	Salesman
P6	Male	25	Hispanic	Bachelor's degree	Research technician
P7	Male	30	Black or African American	Bachelor's degree	Full-stack programmer
P8	Male	50	Black or African American	Graduate degree	Business owner
P9	Female	43	Black or African American	Associate's degree	Self-employed
P10	Female	28	White	Some college (no degree)	Unemployed
P11	Male	48	Black or African American	Bachelor's degree	Engineer
P12	Female	30	Hispanic	Some college (no degree)	Unemployed
P13	Male	47	White	Associate's degree	Media and advertising
P14	Male	50	Black or African American	Bachelor's degree	Artist
P15	Male	29	White	Bachelor's degree	Finances
P16	Female	50	Black or African American	HS Diploma	Consumer service
P17	Non-binary	19	White	Some college (no degree)	College student
P18	Male	29	White	Bachelor's degree	Audio engineer
P19	Male	42	Black or African American	Bachelor's degree	Teacher
P20	Female	54	Black or African American	Graduate degree	Unemployed (disabled due to Covid)

AI memory to human memory? What similarities and differences do you see?"

Following this, participants viewed two short tutorial videos from OpenAI's official website [49]: one showing how to view and manage GPT's memory and another demonstrating how to turn the Memory feature on and off. We then asked, "How have you been managing ChatGPT's memory about you?" and "Do you think ChatGPT can truly forget users' memories?"

Finally, we introduced the risks associated with AI memorization by sharing anecdotal evidence of ChatGPT inadvertently disclosing users' private information to others due to data memorization during training. This example has been used in a previous study [77]. The participants were then asked, "What are your thoughts on using your conversations for training AI models?" and "Are there specific types of data you are concerned about being memorized by ChatGPT?"

We closed our interview with two board questions: "Do you think your life would be better if you turn the memory feature off?" and "What suggestions do you have for ChatGPT designers to better protect users' privacy in relation to the memory feature?"

A detailed interview script is available in the Appendix. Before conducting the interview, we pilot-tested the effectiveness of our interview protocol with three participants recruited through the first and second author's social networks. Based on their feedback, we removed several redundant questions about how participants manage their privacy on ChatGPT and revised the wording of a few questions to improve clarity, such as replacing "algorithmic memory" with "AI memory."

3.4 Qualitative Coding Procedure

The recorded interviews were automatically transcribed by Zoom after each session. We used an open-coding approach to analyze the data, which involves the process of "breaking down, examining, comparing, conceptualizing, and categorizing data" [60]. We began with the first 10 interviews: each author was assigned two transcripts, and the first author reviewed four. All authors were asked to carefully read their assigned transcripts line by line and highlight any text that reflected users' perceptions and experiences with ChatGPT's memory feature, as well as their privacy concerns and the strategies they used to manage AI memorization. The authors were also asked to assign low-level labels to capture the key ideas represented in the highlighted excerpts. These excerpts and their associated codes were then reviewed by another team member for validation. Any discrepancies in coding were discussed and resolved during a cross-validation stage. We addressed discrepancies by revisiting the highlighted excerpts and low-level labels and asked team members to share their interpretations of the participants' comments. We identified areas of shared understanding and re-evaluated the labels that had been assigned.

After this, the first author grouped the low-level codes into broader themes and identified the relevant high-level theoretical concepts (e.g., unforgetfulness, detailedness, accuracy, perceived benefits, perceived risks, call for transparency), linking each concept to their corresponding research question. The resulting categorizations were then shared with the research team for further discussion and refinement. Based on the preliminary coding structure, we repeated the same procedure for the additional 10 interview transcripts to expand and review the coding system.

After completing the coding of the 20 interviews, the research team met again to discuss the findings. Some disagreements emerged due to different understandings of high-level theoretical concepts, such as transparency. When this happened, we discussed the nuanced meanings of the concept and broke the concept into more specific dimensions, such as transparency about the data being stored and transparency about the existence of the memory feature. This helped us clarify our interpretations and resolve disagreements. We document the finalized coding system in the supplementary material.

4 Results

In line with our research questions, we first present participants' perceived similarities and differences between ChatGPT's memory and human memory (RQ1). We then examine instances where participants updated their mental models after learning what ChatGPT remembered about them (RQ2). Finally, we summarize users' suggestions for improving ChatGPT's memory feature to enhance awareness, user experience, and privacy protection (RQ3).

4.1 RQ1: Perceived Differences and Similarities between AI Memory and Human Memory

We found that ChatGPT's memory was perceived differently from human memory on four major dimensions: perceived unforgetfulness, detailedness, accuracy, and the lack of emotions. In contrast, participants also perceived relational benefits from both ChatGPT's memory and human memory.

4.1.1 Perceived Unforgetfulness. Several participants emphasized ChatGPT's ability to retain information over long periods, which is beyond human capabilities. P13 observed, *"It doesn't degrade like human memory."* Similarly, P15 appreciated the persistence of ChatGPT's memory, *"50 years go by, you might forget the name of someone's daughter if you don't see them or that they like jellyfish. That seems like a very minute detail. I don't think ChatGPT has the capability of forgetting if it's already stored and used in your memory feature."*

P19 highlighted unforgetfulness as a key distinction between ChatGPT's memory and human memory, stating, *"At some point, as a human being you can forget…But then there is an AI which remembers."* P17 also believed that ChatGPT has stronger long-term retention, *"Even if I'm gone for a long time, or I delete a chat, it still holds on to that memory. Some humans can do that, but because it's a machine, it probably does do it better."* P14 echoed this point, remarking, *"We are finite, you know, AI is infinite."*

The perceived unforgetfulness of ChatGPT's memory was particularly valued by participants with memory challenges. For example, P20, who took over 30 medications and used ChatGPT to support her disability, found ChatGPT's memory feature very helpful, *"For the medications, I start the chat by saying this is a prescription and ChatGPT memorizes. And when I tell ChatGPT to pull up prescriptions, it'll bring up my list."* However, she also recognized the downside of ChatGPT not able to forget: *"We have the luxury of forgetfulness, and AI doesn't forget anything."* As she explained, this permanence can shift the burden back to users, *"It does not forget, so it kind of holds us accountable or responsible for what we have said or discussed."*

At the same time, a few participants questioned whether ChatGPT should memorize basic details. P19 explained: *"I'd love GPT to hold on to a memory that I'll forget someday. But about my name, I'll never forget my name, I'll never forget where I come from. I'll never forget where I'm living currently, so I don't think it has any good to be stored by ChatGPT. The only thing that I'll be happy to store is something that I can forget."*

However, believing in ChatGPT's unforgetfulness also raised doubts about whether deletion requests truly work. P20 noted, *"It's forgotten to me, but I think it is still used by ChatGPT in some forms."* P14 shared a similar concern: *"Obviously, it's possible that AI can forget, but the question is, is it done or not? I don't know."* Similarly, P3 did not believe that deleting chat histories would do anything to protect privacy, *"I have a feeling that they still have it in the database somewhere. Even if I delete it, I'm just deleting it from my computer."*

Another negative outcome of ChatGPT's inability to forget is that stored information can become outdated and therefore irrelevant to the current task. For example, P10 shared her story about how an old kept memory had influenced ChatGPT's responses to her recent questions, *"It kept one of my memories that we were late on our rent, and from then on it always seemed to assume we were late every month. I needed help writing a text to our landlord, just normal communication, nothing to do with rent, but it would sneak in something about late rent."* P8 had a similar experience, stating *"There's a lot of stuff here about projects that I've already completed. I could definitely delete some stuff in here. I don't need it to remember anymore."* P9 shared her personal information with ChatGPT during her last usage, and she found that ChatGPT kept her old information, recalling *"she is separated."* P9 commented, *"I will have to go in and update it and let them know that I'm not."*

By contrast, a small group of participants believed that ChatGPT can forget. For example, P17, who primarily used ChatGPT to create storylines, remarked that *"sometimes it'll forget some details, and you have to remind it."* P11 added, *"When AI memories are full, it won't be able to save new information."*

4.1.2 Perceived Detailedness. Perceived detailedness emerged as another dimension that distinguishes ChatGPT's memory from human memory. Similar to the varied perceptions of unforgetfulness, participants expressed different perceptions of the detailedness of ChatGPT's memory. One group of participants was impressed by ChatGPT's detailed and vivid memories. As P12 commented, *"It's almost like it's writing an essay about me."* P3 noted that ChatGPT remembered her birthday, birth time, and the place where she was born. After reading a list of memories ChatGPT had about her, P20 felt ChatGPT's memory was *"exactly like human memory. It's very specific."* P11 commented that, *"It (ChatGPT) has the capacity to capture more detail and to remember it effectively."*

P17 highlighted the importance of having ChatGPT memorize all his interactions, *"Everything that I do with ChatGPT, I need it to be memorized."* Based on his interaction with ChatGPT, P17 believed that ChatGPT's memory seemed *"more designed to keep bigger chunks of important, relevant information"* and described ChatGPT as *"a little more finicky with fine details."*

4.1.3 Perceived Accuracy. Furthermore, the accuracy of ChatGPT's memory emerged as an important difference from human memory. For example, P13 commented, *"it (ChatGPT) doesn't subjectivize it."*

He further explained, *"ChatGPT memory is more like a camera or like a transcription of what happened, it can't necessarily judge what happened.... It's not going to be as chaotic as the way humans remember things."* However, one participant (P18) noted that ChatGPT actively learned from users, potentially leading to less accurate and objective responses. As P18 noted, *"With long-term memory, it might change the way it interacts with you. You won't get like an objective answer when you're seeking an objective answer, because it knows certain things about you and how you interact with it."*

4.1.4 Lack of Emotions. In addition, two participants noted the lack of emotions in ChatGPT's memory. For example, P3 described ChatGPT's memory as being different than human memory, remarking, *"it must be like an algorithmic memory."* She further explained, *"I think (human) memories are often hazy and colored by emotions, whereas it (ChatGPT's memory) is probably remembering everything in terms of the codes that's plugged in, like on a scale of 1 to 10, ChatGPT believed that I like formality on a 7."* P12 echoed this view, stating, even though ChatGPT has memory, *"it's different than a friend in that it really can't be comforting. It lacks that comforting or empathy ability."*

4.1.5 Perceived Relational Benefits. Despite the differences, participants also noted similarities between ChatGPT's memory and human memory, particularly in their relational benefits. P3 often used ChatGPT for work, *"memory sometimes makes it feel a little like a person, especially since it will address you by name."* P17 observed that ChatGPT remembered her hobbies, which he found *cute* because it resembled the way humans recall details to bond: *"It just kind of reminds me more of how a normal human would try to remember what you'd like to bond better with you."* P12 appreciated ChatGPT's memory feature for its emotional support, explaining: *"When I have ChatGPT up, it's like there's someone there for me that remembers."* Similarly, P19 compared it to friendship, stating: *"When you have a friend, a person has always had memory. If you went to the beach last weekend, you don't have to remind this friend that last weekend you were at the beach. This person remembers that. So this feature gives me a sense of belonging."*

4.2 RQ2: Mental Model Shifted After Knowing What ChatGPT Remembered

Throughout the interviews, most participants experienced shifts in their mental models after knowing more about what ChatGPT remembered about them. These changes helped participants develop a more accurate understanding of the memory feature, but they also created confusion and uncertainty about how ChatGPT handled data storage, retrieval, management, and privacy protection.

4.2.1 Negative Expectancy Violations. Most participants described experiencing negative expectancy violations, primarily because ChatGPT retained unexpected memories, made inaccurate inferences, or confused highly literate users due to limited transparency.

One common source of these negative violations was due to ChatGPT retaining information that participants did not expect it to remember. For example, P10, who was currently unemployed and actively learning new skills to transition into a user experience designer role, was surprised to find that ChatGPT had stored her *"birth data down to the exact minute and location."* Although she had previously shared personal information openly, explaining that she had experienced *"so many data breaches in the past"* and felt her information was *"already out there,"* discovering the extent of what ChatGPT remembered prompted her to reconsider her privacy management practices. After learning that she could disable data sharing, P10 ultimately chose to turn that feature off.

Likewise, P11 assumed that ChatGPT only retained basic information about him, such as his demographics and occupation. When he reviewed the stored memories, he was shocked to discover that ChatGPT also knew his exact age—information he did not expect the system to hold with such accuracy. He described this as *"a little bit intrusive"* and expressed a desire to delete it. This discovery motivated P11 to delete all personal information from ChatGPT and create a new profile, explaining, *"this time I'll be more careful with the information that I'll be sharing with the chatbot."* He noted that the purpose of this new profile was *"to have a bit more anonymity."*

P3's negative experience with ChatGPT's memory feature was triggered by both unexpected memories and inaccurate inferences made by ChatGPT. P3 primarily used ChatGPT for tasks like resume editing and email writing. She had accidentally cleared out all the memories on ChatGPT and started accumulating new memories a week ago. However, ChatGPT kept her cat's information. She thought that information should have already been deleted, *"I guess it must have saved something if it remembers my cat."* Down the list, P3, whose first language is not English, noticed a specific AI memory regarding her writing style, "She prefers casual tones." However, P3 felt it misrepresented her intentions: *"When I type to colleagues or even to friends, I try very hard to type correctly. But when I type to ChatGPT, my grammar isn't correct, and ChatGPT is obviously processing what I write. It is interpreting my language (preference) incorrectly, like it is accounting for every misspelling and all my grammar that doesn't come out correctly."*

In addition, P20 was dissatisfied with ChatGPT's memory feature, primarily because she felt the system generated an undesirable image of her. *"If you look at everything that ChatGPT remembers about me, I don't know if I like the image that it portrays."* When asked what kind of image she wanted to leave on ChatGPT, P20 responded, *"more professional, maybe."* The mismatch between the AI-generated profile and the desired user profile may have put P20 off and led her to rethink the value of using the memory feature.

Another major source of negative expectancy violations is the lack of transparency in how the memory feature works across models. For example, P1 was a college student majoring in computer science. He was always ahead of the curve when using more advanced models from OpenAI. When engaging in the study activity to identify what ChatGPT had memorized about himself as a user, he immediately asked which model to use and pointed out the response differences between different models, *"o1 Mini's gonna give some extra additional side context, and it'll help me sort of understand what the robot's thinking... When you use 4o by itself, you're just going to get a direct answer."* P1 first tried the o1 Mini model, but it returned with no memories about him. When he switched to the 4o model, it recalled his preferred response style, location, timezone, and the current date and time. He questionned the o1 Mini model, *"What information is available to you?"* The model responded, *"I do not have any information about you."* This unsettled him. He then typed, *"You don't even have my IP address?"* and the o1 Mini replied, *"That's correct. I don't have access to your IP address or any other personal information."* P1 was shocked: *"The fact that one model has an IP address and the other doesn't…I can't explain this… This is just unacceptable."* He continued, *"That lack of transparency and that lack of apparentness leads me to feel like I'm now using a website that I thought that I understood better than I actually do, and that worries me."* He concluded, *"This one (the o1 Mini model) has access to that data, but it's not telling me."*

4.2.2 Positive Expectancy Violation. Positive expectancy violations occur when reality exceeds expectations in a positive direction. Participants were surprised by ChatGPT's accurate information recall and its ability to learn from users. For example, P5 believed that ChatGPT had *"everybody's data"* and worked like a *"master structuring,"* so it did not have detailed data for *"a specific person."* However, after prompting ChatGPT to answer "What do you know about me?" P5 was shocked by its inference capability and the summary ChatGPT provided, *"It referenced my past work experience and seemed to combine it with my prompts. That's why it's blowing my mind. It's talking about character traits and decision making at the same time."* P5 expressed surprise at its accurate recall of his career goals, remarking, *"I didn't know it was gonna be that accurate. That's crazy."*

P3 was also surprised to discover that ChatGPT had remembered her preferred tones for email writing. She remarked, *"Oh, look! It memorized that time I asked it to fix an email and stop sounding so much like business jargon. That was really funny. It was learning from me."* P3 explained that she appreciated ChatGPT's inference capability because *"it makes it easier to work with it."* P3 also preferred keeping all the memories, noting that *"it'll maybe extrapolate something from it later that could help it."*

4.2.3 Positive expectancy confirmation. Positive expectancy confirmation refers to the situations when anticipated positive outcomes are realized. P4 was a part-time software developer and had primarily used ChatGPT for coding. She was aware that *"computers can be hacked"* and *"personal information can be gleaned,"* so she was skeptical and often avoided *"giving personal information that can be misused by somebody else."* In addition, she was well-equipped with the knowledge to protect her privacy on the Internet, such as using VPN to disguise her true IP address and knowing how to turn off ChatGPT's memory feature. Probably due to high AI literacy, she was very careful of what she shared with ChatGPT and often *"tweaked it"* to get the information she needed. She felt reassured after learning that ChatGPT did not have much information about her.

4.2.4 Negative Expectancy Confirmation. Negative expectancy confirmation occurs when unfavorable expectancy meets undesirable reality. P19 read out loud the memories ChatGPT had about himself, and he was surprised that the system had memories of his girlfriend, *"I've come to realize it has my name, my location, and about my girlfriend."* Previously, P19 worried that ChatGPT knew his personal information, *"It was my worst fear, and I've confirmed that it has that information."*

4.3 RQ3: Users' Suggestions to Improve ChatGPT's Memory Feature

Our participants offered four primary suggestions to improve ChatGPT"s memory feature, including 1) increasing visibility and accessibility of the feature, 2) enhancing the transparency of the memory feature in data storage, retrieval, and sharing, 3) restoring user control in the use of the memory feature, and 4) preventing privacy risks from AI memorization.

4.3.1 Increasing Visibility and Accessibility of Memory Features. Several participants expressed concerns about low visibility and limited accessibility of ChatGPT's memory feature. For example, P13 commented, *"I didn't know it (ChatGPT's memory feature) was on this whole time. I'm assuming it was. I don't know how long it's been around."* Similarly, P6 noted, *"I didn't know you can take out the memory or edit the memory."* P15 echoed this sentiment, *"I know that it (information about ChatGPT's memory feature) is available on their website, but I don't know how many people know that this exists. For example, like myself, I didn't know the true ins and outs, that it actually stores specific details and that you can delete it."* Furthermore, P18 highlighted that the memory feature is not easy to locate, *"you have to click four times to see your memory feature. It should be more prominent."*

P18 suggested, *"Having some sort of disclaimer that shows up, maybe not every time you open it, but every few times, just so everyone is aware that your memory feature is on."* P15 shared a similar perspective, *"It would be nice to have some sort of disclaimer more readily available than just in the back end or terms and agreement that people don't really read. I would say tell us what exactly it stores for how long, where it's stored, things of that nature."*

In addition, P15 recommended incorporating multiple formats when designing the disclaimer, *"including a video of this on the main website or a link like 'Are you curious about ChatGPT's memory features?' when you log in. I'm not sure if that already exists or not, but I would like that as a feature, something that's easily accessible."*

P19 also emphasized the importance of informed consent, *"Let a person be informed that there is this feature. Let the person give consent that he or she is ready to use this software with this feature."* P6 appreciated the educational content he received during the interview and expressed interest in more resources about *"how memorization feature works, and how you can leverage it or protect yourself also through it."*

4.3.2 Enhancing Memory Features' Transparency in Data Storage, Retrieval, and Sharing. Another need participants expressed was greater transparency in how the memory feature works, as they were uncertain about what information was stored and whether the deleted information truly became inaccessible. P4 reflected this concern, explaining that even with the memory feature turned off, she *"cannot be 100% certain if ChatGPT is retaining users' information."* She added: *"How do we get a proof that it has actually removed everything? if designers can come up with a design where the user can have some sort of actual proof, not just saying or writing something that 'Yes, all the memory is removed from this app,' that would be helpful. But I do not know whether that's going to be possible."*

P11 shared similar doubt, admitting, *"I am not quite sure how the data is processed and stored… and what measures they have* *taken to protect this information from any data, breaches, or security vulnerabilities."* Beyond security risks, P11 also worried about the potential *"misuse of the stored information,"* especially when it involved personal details.

Some participants also proposed solutions. P6 suggested that *"even just being transparent about how and when this data is used, can help people even take more responsibility and actually even seek out using these features."* Likewise, P16 emphasized the importance of proactive disclosure, noting that she would like to receive *"notifications or disclosures of where that data is being stored and how it is being used."* In addition, P11 suggested implementing a real-time memory indicator: *"That shows you when ChatGPT is adding something to its memory, so you'll be able to know which information it has stored."*

4.3.3 Restoring User Control. User control emerged as another area for improvement. Several participants felt they had some control when using the memory feature. As P2 commented, *"It makes me feel like 'Men in Black,' where you can select your time frame of what you remember, and switch the button, and certain things the person still remembers, and it's like that selective memory."* Similarly, P11 appreciated the feature, saying, *"You can decide whether to turn it off. When you feel the need to have it off, you have the power to do that."*

However, some participants suggested that the memory feature could be further improved by providing more user control. For example, P8 commented, *"It'd be cool if it asks you what kind of things you want ChatGPT to remember."* P17 echoed this point, saying, *"It would be very useful to decide the level of what it remembers. Maybe also allow people to insert their own memories or correct incorrect memories."* Experienced users like P1 emphasized the need for even stricter control: "*Stop memorizing my stuff and stop using it unless I tell you to."*

As for suggestions, P1 proposed an *"auto-categorization feature"* that could be customized by users. He elaborated, *"it should allow users to categorize and manage stored memories, like you can categorize them as work-related and personal files when interacting with the chatbot."* In addition, two participants pointed out that adding time limits to the memory feature could increase user control. P7 explained, *"When it says memory updated, maybe there could be a little drop-down, where you can say memory for 90 mins or 60 mins, something that only remembers for a nice short period of time."* P5 supported this approach, suggesting, *"Instead of just turning it completely off, they could have something like 'erase after 6 months' or 'after 3 months.' Something like that will probably help out a lot."*

4.3.4 Preventing Privacy Risks from AI Memorization through Alerts and Intervention. In addition, our participants expected ChatGPT's memory feature to alert them about the potential privacy risks associated with AI memorization. As P7 pointed out, *"it could tell you not to put certain information, or it redacts it."* Similarly, P6 suggested that having an AI warning that flags sensitive information would be useful, e.g., *"We've detected sensitive details about you based on this conversation. Would you like to delete the conversation afterwards or prevent it from being used in other ways?"* P8 emphasized the need for more restrictions regarding the sharing of other people's information: *"if you ask for someone's phone number or email address, it should say, 'We don't give that out.'"*

5 Discussion

This study examines users' perceptions and experiences with ChatGPT's memory feature in relation to their privacy concerns. Drawing on participants' feedback and theoretical perspectives on privacy and personalization, we first discuss the primary characteristics identified by our participants that differentiate ChatGPT's memory from human memory. In addition, we outline major theoretical implications and design suggestions to support transparent, user-aligned ChatGPT memory interactions.

5.1 Machine-like Nature of ChatGPT's Memory and User Trust

First, we identified four major characteristics that differentiate ChatGPT's memory from human memory based on users' perceptions: perceived unforgetfulness, detailedness, accuracy, and lack of emotions. These perceptions highlight the machine-like nature of ChatGPT's memory, which may trigger either the positive or negative machine heuristics. The positive machine heuristic can strengthen the belief that ChatGPT's memory is more objective and neutral than human memory [62, 63], whereas the negative machine heuristic may lead users to view ChatGPT's memory as rigid, mechanistic, and lacking human intuition and subjectivity [45]. In either case, these perceptions can influence users' trust in the system [45, 61, 63].

In particular, the unforgetfulness of ChatGPT's memory elicited contradictory viewpoints. On the one hand, it empowered users with memory challenges to have ChatGPT remember information for them. On the other hand, some participants perceived that the feature produced outdated and irrelevant memories and imposed accountability burdens on users. These findings point to moments when user agency and machine agency are either aligned or in conflict. When ChatGPT's memory serves an assistive role, it seems to enhance user agency by allowing individuals to delegate memory tasks and benefit from mental offloading.

Notably, prior research on Google has shown a downside when information is stored outside ourselves: people tend to recall less of the information itself while maintaining stronger memory for where to retrieve it [59]. While users can search their history relatively easily on platforms like Google, doing so in ChatGPT is more difficult because the current memory feature lacks effective search functions and can reach capacity, limiting the storage of new information. These constraints may ultimately undermine users' trust in ChatGPT and its memory feature.

Building on these challenges, the most critical issue is that participants do not seem to believe that ChatGPT can forget. The concern is not only technical, i.e., whether the system truly deletes data, but also deeply experiential, influencing how users feel and behave. It may reinforce skepticism toward the forgetting mechanism, create reluctance or hesitation to engage with the system, and even generate anxiety or uneasiness about being continuously monitored, recorded, or having their personal information permanently stored. Future studies should investigate how to better communicate the forgetting mechanism in ChatGPT's memory, so that it does not undermine user agency and trust in the system.

5.2 Relational Perceptions of ChatGPT's Memory

Despite viewing ChatGPT's memory as machine-like, participants repeatedly linked ChatGPT's memory to human memory in relational terms. They used words and phrases, such as "friendship" and "sense of belonging," to describe how they related to AI memory interpersonally. This finding highlights that ChatGPT's memory can evoke social responses from users [47]. It functions not only as a technical mechanism (e.g., storing or recalling data), but also as a social one, prompting users to apply social rules, treat the tool as a friend, and form relational understandings through the system.

Designers could deliberately scaffold this relational opportunity by offering customization controls that mediate between "utility" (task memory) and "affect" (relational memory). In practice, this means giving users the ability to decide what kinds of memories the system should prioritize and how they should be expressed in interaction. However, a critical question is how much "friendship" should be engineered into AI's memory. Although relational cues can strengthen perceived rapport, if they are too strong or too persistent, they risk fostering over-trust [61] or creating a false sense of intimacy that makes users more vulnerable when the system errs or spreads misinformation [40]. Designers should balance utility and affect, ensuring that ChatGPT's memory supports practical tasks without misleading users about the system's capabilities or intentions.

5.3 Mitigating Expectancy Violations in Users' Experience with ChatGPT's Memory Feature

In addition, our participants' reactions mapped closely to the theory of expectancy violations [10]: most experienced negative violations, while others had positive surprises. The primary causes of negative reactions were ChatGPT's memory feature recalling unexpected memories, making inaccurate inferences, and a lack of transparency and explainability when different models provided different memories. Notably, participants with higher AI literacy were more likely to experience either positive expectancy confirmations or negative expectancy violations. On the one hand, their strong privacy management strategies, such as limiting the personal information they shared, often resulted in ChatGPT retaining very little about them, which aligned with their expectations. On the other hand, these participants also encountered more negative expectancy violations when ChatGPT's memory behaved differently than they expected. When their mental models were challenged, they experienced heightened frustration and disappointment because ChatGPT's memory did not operate in the way they anticipated.

To mitigate negative expectancy violations, ChatGPT could adopt explicit disclosures of data management early on, thereby forming a more accurate mental model among its users. This early communication is likely to shape positive perceptions and trust in the system, as evident in previous studies [12, 45]. In addition, the lens of expectancy violations emphasizes that transparency is not just about disclosure, but also about predictability. When the retained information does not match users' expectations, they seem to experience greater frustration and confusion. Interfaces need to make memory actions legible and auditable in real-time, for example, through live indicators. This extends prior HCI work on algorithmic transparency [36, 41, 58, 75] by foregrounding memory as a new trust boundary.

5.4 Privacy Risks in AI Memorization and User Profiling

Previous studies have highlighted different types of privacy risks related to AI memorization, such as personal data leakage [33, 77], sensitive-attribute inference [57], and risks amplified by third-party integrations such as API [57]. In our interview study, two types of privacy concerns emerged more frequently than others. The first concern centers around the perception that ChatGPT's memories are highly detailed and retentive. This often leads to unexpected resurfacing of information that users themselves may not remember sharing, or that may no longer be relevant to the current context. The system's capability to retain detailed coverage of extensive user information can become a privacy issue because users lack control over how their information is stored and retrieved, raising suspicions about how ChatGPT might use their data in future interactions or for other purposes. This concern is further intensified given that ChatGPT has shifted from a nonprofit organization to a commercial one [26].

In addition, our interview data revealed a new category of privacy concerns: undesirable user profiling. User profiling refers to the process of gathering and analyzing data about a user to create a detailed profile of their behavior, preferences, and characteristics, and it is enabled by RAG-based memories, which extract and store user information across sessions for later use by the system. While adaptive user profiling based on memory is not new [67], a key challenge remains: the profile created by LLMs may not align with the image users desire to project. This misalignment heightens tensions between machine agency and user agency, specifically over which image should be retained and who ultimately benefits from the resulting profile.

Adding to this concern, the relational benefits of AI memory, such as increased continuity, personalization, and a sense of being "known" by the system, may further complicate the power of AI-generated profiles. As users grow accustomed to more fluid and context-aware interactions, they may become less likely to question, scrutinize, or correct the profiles the system constructs about them. This relational ease can mask underlying inaccuracies or biases in the profile, allowing the system's interpretation of the user to become more influential than the user's own self-presentation.

5.5 Design Suggestions

By considering participants' concrete suggestions and the discussions emerging from their comments, we provide several design suggestions that emphasize transparency, accessibility, agency, bounded friendship, and scrutiny as core components of creating a more responsible, user-aligned memory experience. We further draw on the theoretical frameworks of privacy calculus and privacy-personalization paradox to discuss how AI memory systems can support dynamic trade-off negotiation. Table 2 summarizes the design suggestions informed by users' comments and theories of privacy and personalization.

5.5.1 Increasing Transparency of Memory Existence and Use. A fundamental issue was that many participants were unaware that

ChatGPT's memory feature existed at all, confirming earlier findings on users' low memory awareness and literacy [54]. Therefore, transparency needs to begin with clear onboarding prompts that inform users that memory is available, describe its purpose, and allow them to decide whether to opt in or not. Moreover, users' consent should be requested whenever memory updates occur. This consent mechanism should be applied universally, rather than only in regions such as the UK and the EU, where it is currently required [50].

Transparency also involves revealing how memory affects interactions. Our participants wanted to understand which past details were resurfaced, why certain information was retained, and what happens after deletion. Providing concise and easily accessible explanations, potentially in multiple formats, such as text or video, could make these processes more understandable without overwhelming users with technical jargon. Additionally, providing situational cues, such as showing which memory snippet is currently informing a response, would allow users to understand the relevance of specific details within context. These design solutions support a more interactive and reflective relationship with memory, enabling users to identify, access, and correct potential privacy risks as they arise.

5.5.2 Improving Access and Organization of the Stored Memories. Participants consistently described the current memory feature as hard to locate and unintuitive. Locating memory within mutli-level settings menus not only discourages users from checking what has been stored but also limits opportunities to correct or organize their data. A more accessible design would position memory as a first-class interface element, for instance, in the main navigation panel, where users already browse chats and tools. Such placement could normalize memory as an active resource rather than a hidden backend process. Beyond access, participants emphasized the need for more structured organization of stored memories. Categorizing memories into work-related, personal, or user-defined groupings could help users navigate and manage information more effectively.

5.5.3 Strengthening User Agency Through Editable and User-Shaped Memories. Participants frequently expressed a desire for great agency over what ChatGPT remembers, how long those memories persist, and how those memories shape the system's impression of them. Although OpenAI distinguishes between "saved memories" and "reference chat history" to give users some control [51], participants felt this was insufficient because the system still infers and stores new details automatically, often in ways they cannot anticipate or track.

They emphasized the need for full visibility of all stored information and the ability to edit or delete entries. Some also proposed time-limited retention, allowing certain memories to expire after a user-defined period to align with changing goals or sensitivities. Such temporal flexibility would reduce the risk of inadvertently outdated representations.

Participants also drew attention to the impact of stored memories on their perceived identity within the system. Unwanted or inaccurate profiles can arise when the system overgeneralizes from prior interactions or misinterprets temporary interests as stable characteristics. To mitigate this, the system could initiate a sign-up prompt asking users to define how they would like the AI to perceive them. Expanding upon the current "custom instructions" feature, the system could support multiple context-specific identities, such as professional roles or personal interests, and allow users to switch between them. By treating profiles as collaborative artifacts, designers can avoid privileging system inference over user intention and preserve user agency in self-presentation.

5.5.4 Designing Bounded Friendship and Built-in Scrutiny. Our findings show that participants appreciated the relational benefits brought by ChatGPT's memory feature. To support this, designers can engineer "friendship" into LLM systems through relational cues that draw on memory in subtle, bounded ways. For instance, the system might greet users with light continuity, "Welcome back! Ready to continue your proposal?" Similarly, the interface could offer supportive but non-intimate emotional alignment, such as "That update sounded stressful; let's break it down together."

At the same time, these relational benefits can introduce new risks if left entirely to the system. Without users' involvement, relational comfort may conceal inaccuracies or biases in the AI-generated profile. For example, a non-native English speaker who makes occasional grammatical errors might be mistakenly profiled by the system as preferring a casual writing style; as a result, the AI continues producing informal, loosely structured text even when she is preparing professional documents, ultimately hindering rather than helping her. To counter this tendency, LLM systems should make their inferences about users transparent and revisable. For example, periodically showing a short "profile summary" (e.g., "I've inferred that you prefer a casual writing style") and explicitly enabling users to confirm, correct, or delete these assumptions. Proactively inviting user scrutiny of such inferences can surface and correct inaccuracies in the AI-generated profile before they accumulate and negatively affect future interactions.

5.5.5 Navigating the Privacy-Personalization Trade-offs in AI Memory Interaction. Our findings show that users do not make a single, overall evaluation of benefits and risks when interacting with ChatGPT's memory feature. Instead, managing ChatGPT's memory is an ongoing process in which users continually navigate a shifting boundary between the relational benefits of being remembered and the potential risks of being profiled, misrepresented, or overexposed. This dynamic trade-off requires system and memory designs that support continuous recalibration rather than relying on one-time consent or static settings.

To support this ongoing balancing act, the system should also allow users to adjust the specific qualities of memory that matter most in their experiences, such as its detailedness, its ability to forget, its emotional tone, and the relational benefits it produces. By giving users the ability to adjust these characteristics, it enables users to maintain a memory experience that aligns with their preferences while they continually negotiate the trade-off between privacy and usefulness.

5.6 Limitations and Future Studies

The interview findings should be interpreted with the following limitations in mind. First, this study focused on ChatGPT's memory feature, as it is one of the most widely used generative AI systems on the market [1]. Therefore, the insights may not be generalizable

Table 2: Summary of Key Findings and Design Suggestions

Users' Needs	Example Users' quotes	Design Implications
Need for Transparency of Memory Existence and Use	• "I didn't know it (ChatGPT's memory feature) was on this whole time." • "I didn't know you can take out the memory or edit the memory."	• Clear on boarding with the memory feature • Ask users whether they want to opt in • Present a disclaimer to increase awareness of the memory feature • Provide informed consent • Request users' consent when memory updates its policy and functions
Need for Easy Access and Improved Organization of the Stored Memories	• "You have to click four times to see your memory feature." • "It should allow users to categorize and manage stored memories."	• Position memory as a first-class interface element in the main navigation panel • Provide an auto-categorization feature • Implement time-limited retention
Need for Transparency in Data Storage, Retrieval, and Sharing	• "I am not quite sure how the data is processed and stored." • "It's possible that AI can forget, but the question is, is it done or not? I don't know."	• Transparent about when and how the data is used • Provide notifications or disclosures of data storage and usage • Provide concise and easy-to-understand explanations of how memory influences users' interactions • Provide situational cues to explain how memory informs a specific details within context
Need for User Agency Through Editable and User-Shaped Memories	• "It would be very useful to decide the level of what it remembers." • "Stop memorizing my stuff and stop using it unless I tell you to." • "If you look at everything that ChatGPT remembers about me, I don't know if I like the image that it portrays."	• Allow users to insert their own memories and correct existing entries • Implement real-time memory indicator • Initiate a sign-up prompt asking users to define their desired profiles • Support multiple context-specific identities through the "Custom instructions" feature
Need for Prevention of Privacy Risks from AI Memorization	• "It had my birth data down to the exact minute and location."	• Redact personal or sensitive input • Flag sensitive information through system warning • Restrict the sharing of other individuals' information
Need for Bounded Friendship and Built-in Scrutiny	• "When I have ChatGPT up, it's like there's someone there for me that remembers." • "Memory sometimes makes it feel a little like a person, especially since it will address you by name."	• Engineer "Friendship" into LLM agents through the presentation of relational cues • Show a short profile summary periodically • Invite users to confirm, correct, and delete these assumptions

to other LLMs and generative AI systems that implement memory features differently.

In addition, memory features are constantly evolving. As a one-time study, this work only captures how users perceive ChatGPT's memory feature at a specific moment in time. With increased experience and knowledge of LLM-based memories, these perceptions may change. As OpenAI and other platforms continue to update their memory features, it would be valuable to document these changes and examine how users' perceptions of AI memories evolve over time.

Third, our participants were recruited from Prolific and limited to individuals currently living in the U.S. Given the variation in cultural attitudes toward privacy and AI [7, 70], our findings may not be generalizable to users from other cultural backgrounds. Relatedly, we intentionally recruited current ChatGPT users to understand their perceptions and experiences with the memory feature. This approach may have excluded individuals who avoid the platform due to privacy concerns, potentially omitting important viewpoints relevant to the issues under study. Future research should recruit a more diverse sample and include individuals who do not use

ChatGPT to ensure a more comprehensive investigation of users' perceptions of memory features.

Another major concern is the use of anthropomorphic terms such as "remember" and "forget" when describing ChatGPT and its memory feature throughout the paper. These word choices mirror the languages embedded in the current design of ChatGPT's memory feature and were also used intentionally by our participants to express their perceptions and experiences with the feature during the interviews. However, it is important to recognize the ethical and epistemological risks associated with anthropomorphizing AI and its ability. For example, Placani [52] argued that attributing human-like traits to AI can inflate perceptions of its capabilities and performance. Anthropomorphic language can also distort moral judgments about AI, shaping views of its moral character, status, responsibility, and trustworthiness [52]. In addition, Salles et al. [55] pointed out that anthropomorphizing AI may fuel both exaggerated fears about AI and unwarranted optimism. Future studies should examine how anthropomorphizing AI's memory and forgetting capabilities influence users' cognition, affect, and behavior toward LLM agents and consider replacing anthropomorphic terms with neutral ones to reduce the negative consequences on users.

Last, LLMs employ different types of memory, such as contextual memory, RAG-based memory, and epistemological memory. Since lay users often struggle to distinguish among these types of AI memory [76], and the study focuses on users' holistic experience with ChatGPT's memory feature rather than a specific technical memory mechanism, we did not differentiate types of AI memory in the interviews. Considering these memory types can differ significantly in their privacy implications [76], future work could benefit from clearly examining specific AI memory types and providing memory-specific privacy recommendations.

6 Conclusion

This study examines users' perceptions and experiences with ChatGPT's memory feature. Our interview findings show that participants perceive both relational benefits and privacy risks from their engagement with the feature. By grounding these insights in the theoretical frameworks of privacy calculus and privacy personalization paradox, we highlight the importance of viewing human-memory interaction as a dynamic process that requires ongoing negotiation, shaped by how detailed, unforgetful, accurate, and relational the system appears in practice. By illuminating these tensions, this work deepens our understanding of how users perceive ChatGPT's memory feature and underscores the need for future designs that protect privacy while preserving the relational benefits of personalization.

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A Appendix: Interview Script

A.1 Greeting and Self-Introduction

Hi, nice to meet you! Thank you for joining today's interview session. How are you doing today?

Great. I am a researcher from X university, and I will walk you through the interview today. As we shared earlier in the study description, this study is about ChatGPT's memory feature. My research team and I are interested in understanding users' perceptions and experiences with this feature. Since the conversations will involve sharing your personal experience with ChatGPT, please feel free to share whatever you feel comfortable sharing. The interview will be one hour long, and we will ensure some time for you to ask questions at the end. Does that sound good to you?

A.2 Requesting Permission to Record

Before we get started, I would like to ask whether it is okay to record the session through Zoom. Please note that only members of our research team will have access to the transcribed interview data, and we will keep your responses confidential. Do you have any questions about the recording process?

Can I start recording the session?

(After receiving participants' approval) Thank you. I will start recording now. Let's begin.

A.3 Participant Introduction

First, could you please briefly introduce yourself, including your preferred name, current location, and occupation?

A.4 Use of ChatGPT in General

Great. Let's talk about your use of ChatGPT in general.
- Which version of ChatGPT do you use? Free or paid?
- How long have you been using ChatGPT?
- What do you primarily use ChatGPT for?
- Do you share any personal information with ChatGPT? If so, why?
- How do you think ChatGPT manages, stores, and uses your personal data?
- How do you protect your privacy while using ChatGPT?

Thanks for your response. I would like to introduce ChatGPT's memory feature to ensure that we are on the same page. This feature was developed in February 2024 to give users greater control and make future chats more helpful. When users provide new information, ChatGPT may respond with a visual cue indicating that its memory has been updated.
- Have you ever encountered the memory feature in your usage of ChatGPT?
- What do you like or dislike about the memory feature?
- What types of information do you believe ChatGPT memorizes?
- What memories do you think ChatGPT has about you?

A.5 Experiment with ChatGPT's Memory Feature

Now, I would like to invite you to participate in an experiment with ChatGPT. In this experiment, you will be asked to prompt ChatGPT with the question, "What do you know about me?" Please log in to your ChatGPT account and ask this question. I have several follow-up questions for you after you receive ChatGPT's response. Please let me know when you are ready to proceed.
- Was the result surprising to you?
- What information would you like to keep stored on ChatGPT?
- What information would you prefer to delete from ChatGPT's memory?
- Have you ever tried to delete certain memories from ChatGPT?

A.6 Probing into AI Memory

As you probably know, human memory refers to our ability to store and recall information, and it has traditionally been viewed as a uniquely human capability.
- What does it mean to you when AI has memory?
- How would you compare AI memory to human memory? What similarities and differences do you see?

A.7 Managing ChatGPT's Memory

You may already know that there are multiple ways to control ChatGPT's memory feature. I have two tutorial videos from OpenAI's website [49] that explains how to manage ChatGPT's memory feature through system settings and how to turn the feature on and off, respectively. Let's watch them together.
- How have you been managing GPT's memory about you?
- Do you think ChatGPT can truly forget users' memories?

A.8 Memorization Risk

While ChatGPT's memory feature is designed with good intentions, research shows that it is associated with memorization risk, especially when the model uses user data for future training. For example, prior research [77] documented an instance in which GPT-3 offered detailed private information about the Editor-in-Chief of MIT Technology Review, including his family members, work address, and phone number.
- What are your thoughts on using your conversations for training AI models?
- Are there specific types of data you're concerned about being memorized by ChatGPT?

A.9 Concluding Questions

Thanks for a great conversation so far. I have two final questions for you.

- Do you think your life would be better if you turn the memory feature off on GPT?
- What suggestions do you have for ChatGPT designers to better protect users' privacy?

A.10 Compensation and Thank You

Thank you for sharing your perceptions and experiences with ChatGPT's memory feature in today's session. Do you have any questions?

As an appreciation, you will receive $20 payment through Prolific. Could you please confirm the last three digits of your Prolific ID?

Please feel free to contact me if you have any other questions. Hope you have a great day!